Chapter 1

Latent Space Analysis

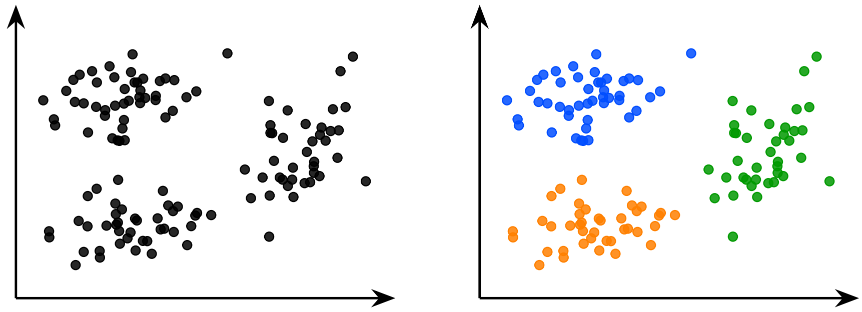
The ability of a neural network to generalize provides evidence that the model has learned more than a simple memorization of the training examples **??**. Instead, it must construct internal representations that capture regularities present in the data. These representations play a central role in the network's ability to solve a task. Understanding their structure is therefore a key step toward understanding how neural networks achieve generalization.

In classical machine learning approaches, practitioners had to manually design data representations in order to make them usable by simple models. Deep learning introduces a paradigm shift: rather than relying on hand-crafted features to meaningfully represent data, it automatically learns these representations. Through the successive activation of hidden layers, neural networks gradually construct representations that become increasingly relevant to the task under consideration [Rumelhart et al., 1986]. As discussed in Chapter **??**, these representations are not meaningful at initialization, they emerge progressively during training as the model optimizes its objective. It is precisely through these learned representations that a trained network is able to map an input to an output, including for examples never encountered during training.

These latent spaces are now at the core of a growing number of practical applications. Neural video compression exploits the compactness of latent spaces to encode information efficiently [Lu et al., 2019]. In reinforcement learning, agent internal representations are used for efficient exploration in extremely open environments [Burda et al., 2018]. Recommendation systems model users and content within a shared latent space to compute meaningful similarities [He et al., 2017]. Similarly, retrieval-augmented generation systems index and query knowledge bases through latent embeddings [Lewis et al., 2021]. Finally, large language models and multimodal architectures rely on the quality of learned representations to align heterogeneous modalities such as text, images, and audio [Radford et al., 2021]. These examples represent only some of the most widespread applications of latent representations.

Given their extensive use, the study of latent representations has become a central topic in deep learning research. A major objective is to understand how information is organized within these spaces, which concepts are encoded, and how these representations influence model behavior.

In this chapter, we first make explicit the working hypothesis that underlies much of the research on latent space analysis. Throughout this manuscript, we



(a) Observed activations **(b)** Under the concept hypothesis

● Concept A ● Concept B ● Concept C

Figure 1.1: The same latent space activations seen without concept hypothesis **(a)** and under the working hypothesis that concepts emerge in the latent space **(b)**. Colors represent hypothetical concept memberships assigned by the model; they are not observed labels. Under this hypothesis the structure of the latent space becomes meaningful and amenable to analysis.

refer to this assumption as the concept hypothesis 1.1. Once this conceptual framework has been established, we review the main approaches proposed in the literature to identify and extract concepts from latent representations. To the best of our knowledge, the main families of methods include probing methods (Section 1.2), semantic directions (Section 1.3), sparse autoencoders (Section 1.4), and clustering-based approaches (Section 1.5).

1.1 Concept Hypothesis

In this section, we introduce the main working hypothesis underlying much of the literature on latent space analysis. Although rarely stated explicitly, this hypothesis is implicitly present in the recurring use of notions such as abstraction, representation, latent space, concept, and semantics. The goal of this section is to make this conceptual framework explicit by formally defining its main components.

This hypothesis originates from the need to explain the generalization ability of deep neural networks observed after training, as discussed in Section **??**. Formally, a model is said to generalize when, for an input $x \in \mathcal{X}$ not seen during training, it still produces an output $\hat{y}$ close to the true output y. Such behavior cannot be explained by simple memorization of training samples. Instead, the model must extract and exploit underlying regularities in the target function. This requires a process of abstraction, as defined in Definition 1.1.1, whereby raw inputs are transformed into internal structures that are meaningful for the task [Bengio et al., 2014].

Definition 1.1.1: Abstraction

Abstraction is the process by which a model transforms an input into a task-relevant internal form. It captures and restructures information in order to represent relationships that are useful for prediction or decision-making.

The ability of neural networks to perform abstraction arises from their intermediate activations. Indeed, a neural network does not directly map x to y. Instead, as illustrated in Figure **??**, it computes a sequence of intermediate activations across hidden layers. These activations correspond to internal encodings of the input, which we refer to as representations in the sense of Definition 1.1.2.

Definition 1.1.2: Representation

A representation is the internal encoding of an input produced by a trained neural network. It corresponds to the activations of one or several layers and provides a transformed view of the input.

Each hidden layer refines the representation of the previous layer. Each transformation progressively reshapes the representation into a form more directly aligned with the output space [Rumelhart et al., 1986]. This process can be described as a sequence of transformations where each $h^{(l)}$ is a representation:

$$x \to h^{(1)} \to h^{(2)} \to \cdots \to h^{(L-1)} \to \hat{y}$$

The set of all such representations produced by a layer or a model over a dataset defines its latent space, as defined in Definition 1.1.3.

Definition 1.1.3: Latent Space

The latent space is the set of all representations produced by a layer or model over a dataset. It is shaped by learning and organizes representations according to task-relevant properties.

Under the concept hypothesis, the analysis of the latent space is assumed to provide access to higher-level abstract structures learned by the model, as illustrated by 1.1. These structures are referred to as concepts, understood here in a philosophically pragmatic sense. In this framing, concepts are not evaluated as true or false ideal entities, but rather as abstract tools whose validity depends on their usefulness for a given task [Newell, 1982]. Our definition is given formally in Definition 1.1.4.

Definition 1.1.4: Concept

A concept is a task-dependent abstraction that captures regularities in the input space relevant for predicting the target output.

This perspective is particularly well aligned with latent space analysis, where the objective is to uncover organizing principles that support the model's behavior. From this perspective, a given representation can be interpreted as a composition of multiple such concepts, each contributing to the final prediction.

Another property that can be considered when analyzing concepts is their relationship to human interpretation. This relationship is captured by the notion

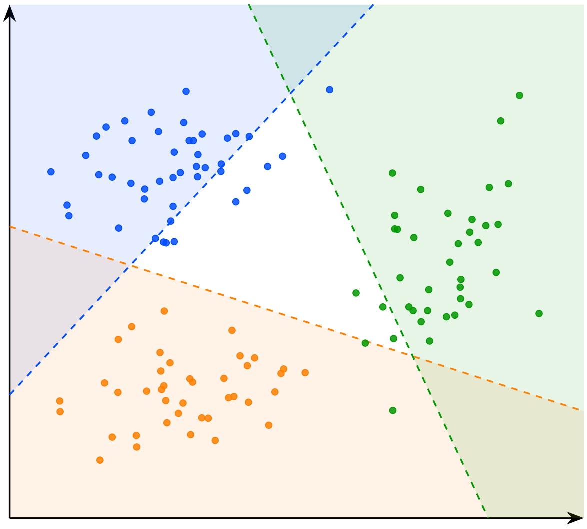


Figure 1.2: Decision boundaries of the linear probes for concepts A, B, and C. The colored regions indicate the areas classified positively by each probe.

of semantics [Marconato et al., 2023], defined in Definition 1.1.5. Under this definition, concepts may exhibit varying degrees of semantic content. Some concepts can be readily associated with human-interpretable notions, whereas others may contribute to task performance while remaining difficult to interpret.

Definition 1.1.5: Semantics

Semantics refers to the alignment between concepts in the model latent space and concepts used by a human observer. A structure is semantic if it can be consistently associated with a human-interpretable concept.

It is important to note that this conceptual framework is not supported by a complete theoretical understanding of deep neural networks. Current mathematical tools do not provide a rigorous characterization of their internal representations and do not formally prove that concepts emerge as identifiable structures within latent spaces. Nevertheless, this perspective constitutes a widely adopted working hypothesis in the field of representation and latent space analysis.

The remainder of this chapter is devoted to reviewing methods for analyzing latent spaces and extracting the concepts they encode. We begin with probing, one of the most direct approaches for assessing whether concepts are represented in the latent space.

1.2 Probing

Probing methods aim to determine whether a given human-defined concept is encoded within the internal representations of a neural network [Li et al., 2024, Karvonen, 2024]. More specifically, they investigate whether the presence or absence of a selected concept can be inferred from embeddings. The underlying

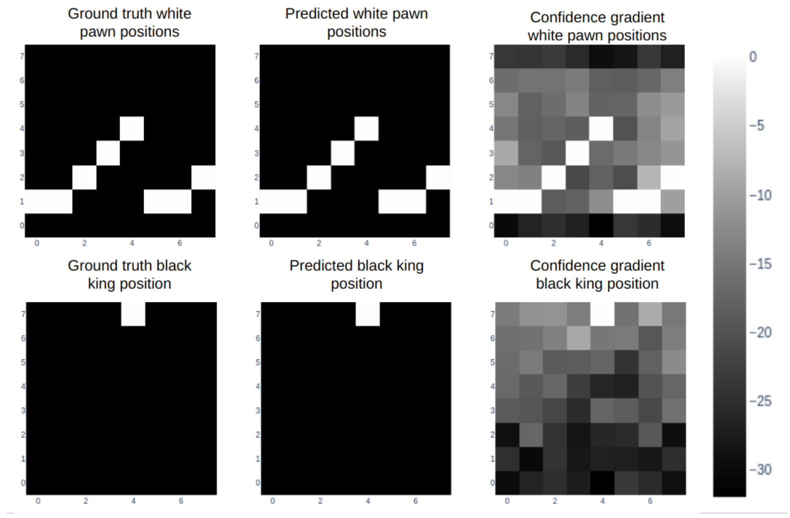


Figure 1.3: Probing analysis of a large language model trained to play chess [Karvonen, 2024]. Each row corresponds to a piece type: white pawns (top) and the black king (bottom). The left column shows the ground truth board positions, the middle column shows the positions predicted by the probe from the model's internal representations, and the right column shows the confidence gradient across board squares. The close agreement between ground truth and predicted positions suggests that the model internally maintains a structured representation of the board state, even though it was only trained on sequences of moves expressed in natural language.

assumption is that if a concept can be reliably decoded from these representations, then the network likely encodes information related to it in order to perform its task.

A probing analysis begins by identifying a set of semantic concepts that the network could potentially use to accomplish its task. Each example in the dataset is then annotated to indicate the presence or absence of the studied concepts. We denote by c_j the presence or absence of concept j in example x. The dataset can therefore be written as follows:

$$\mathcal{D} = \{(x^{(i)}, y^{(i)}, c_1^{(i)}, \dots, c_n^{(i)})\}_{i=1}^{N}.$$

The data are subsequently projected into the latent space of the pretrained model. We denote by h^l the representation vector obtained at layer l. From these latent representations, a classifier is trained in a supervised manner to predict the presence of the considered concept. This classifier is referred to as a probe and is denoted by p. Its purpose is to determine whether a given concept is present in the representation. Its purpose is therefore to assess whether information associated with the concept is accessible from the network's internal representations. This can be formalized as follows:

$$p_\theta^{l,j} : h^l \mapsto c_j, \quad \theta^* = \arg\min_\theta \mathcal{L}\big(p_\theta^{l,j}(h^l), c_j\big)$$

The classifier used to detect the presence or absence of a concept must remain intentionally simple. Indeed, if the probe has too much capacity, it may reconstruct the target information from complex correlations that do not necessarily correspond to information explicitly encoded in the latent space. In such a case, it becomes difficult to determine whether the concept is genuinely represented by the network or merely reconstructed through the expressive power of the classifier. For this reason, probing methods often rely on linear classifiers. A linear separation suggests that the information is easily accessible to the neural network and potentially usable during decision-making. If the classifier achieves performance significantly above chance level, this suggests that the latent space contains exploitable information related to the considered concept.

An application of this technique can be found in studies investigating whether large language models (LLMs) develop internal representations of their environment [Li et al., 2024, Karvonen, 2024]. In these works, researchers train LLMs to play games such as chess or Othello in a purely textual setting. The model is only provided with sequences of moves and information about the actions it can take, all expressed in natural language. The central question is whether the LLM develops an internal representation of the game state. To address this question, the probed concepts correspond to the presence or absence of different game pieces on specific board positions. As illustrated in Figure **??**, probes are able to reconstruct the full board state from the model's hidden representations, even though the only available information to the model is a sequence of text.

However, the fact that a concept can be decoded from latent representations does not necessarily imply that the model actually uses this information to perform its task. To assess whether representations play a causal role in the model's output, it is possible to perform interventions on the latent representations. An intervention consists of modifying a latent representation in a controlled manner by activating or deactivating a given concept, and then studying its impact on the model's output. If the resulting change in the model's output is consistent with the applied perturbation, this suggests that the representation had a causal role.

To this end, we aim to minimally modify the latent representation h^l such that the probe prediction for the target concept c_j changes state, while leaving the predictions associated with other concepts unchanged. This goal can be formulated as an optimization problem over the latent activation itself. RThe optimization is performed directly on the representation $h^{l'}$, using gradient-based optimization as introduced in Section **??**. The solution to this problem is the optimal modified representation h^{l*}, defined as:

$$h^{l*} = \arg\min_{h^{l'}} \underbrace{\gamma\, \mathcal{L}_{\text{flip}}\big(p_\theta^{l,j}(h^{l'}), c_j\big)}_{\text{flip target concept } c_j} + \underbrace{\lambda \sum_{k \neq j} \mathcal{L}_{\text{preserve}}\big(p_\theta^{l,k}(h^{l'}), c_k\big)}_{\text{preserve other concepts } c_{k \neq j}} + \underbrace{\beta \|h^{l'} - h^l\|_2^2}_{\text{minimality}}$$

where:

- h^l is the original latent representation of the input at layer l,
- $h^{l'}$ is the modified latent representation being optimized,
- h^{l*} is the optimal modified representation,
- $\mathcal{L}_{\text{flip}}$: loss encouraging the prediction of the target concept c_j to change state (i.e., flipping the predicted label),

- $\mathcal{L}_{\text{preserve}}$: loss enforcing invariance of all non-target concepts c_k for $k \neq j$,
- $\|h^{l'} - h^l\|_2^2$: regularization term ensuring the modified representation remains close to the original latent representation,
- γ, λ, β: weighting coefficients controlling the trade-off between the intervention strength, the preservation of other concepts, and the minimality of the perturbation.

In the context of LLMs studied previously, such interventions have been used to remove specific pieces from the model's internal representation of the game state [Li et al., 2024, Karvonen, 2024]. The main observation is that, in the vast majority of cases, the model behaves as if the piece had effectively disappeared. In other words, the LLM continues to generate legal moves while correctly accounting for the absence of the removed piece. This provides evidence for the causal role of these internal representations in the model's behavior.

A main limitation of the probing approach is that the concepts under investigation must be predefined before any analysis can be performed. This introduces a strong bias in what can be studied. Moreover, annotating a dataset for each concept is extremely time-consuming and requires significant human effort. Interventions in latent space are also difficult to interpret, as it is not always clear what constitutes a meaningful change in the network's output under a given perturbation.

Probing methods provide a binary view of concepts, focusing on whether a given concept is present or absent in a latent representation. However, this binary perspective can be overly restrictive, as concepts may be represented in a more continuous or graded manner, with varying degrees of presence. To obtain a more detailed characterization of these representations, other approaches have been proposed, such as the analysis of interpretable directions in latent spaces.

1.3 Interpretable Directions

To overcome the limitations of binary concept detection in probing methods, several approaches have been proposed to analyze the geometric structure of latent representations. Among them, the notion of interpretable directions has emerged as a natural extension, aiming to capture the continuous nature of how concepts are encoded in representation spaces. Empirically, it has been observed that shifting a latent representation along a direction can induce coherent and semantically meaningful changes in the model's output. Such transformations can be expressed as:

$$h' = h + \alpha \mathbf{v}_i,$$

where:

- h denotes the original embedding,
- $\mathbf{v}_i$ is a unit vector associated with concept c_i,
- α controls the strength of the intervention,
- h' is the resulting perturbed representation.

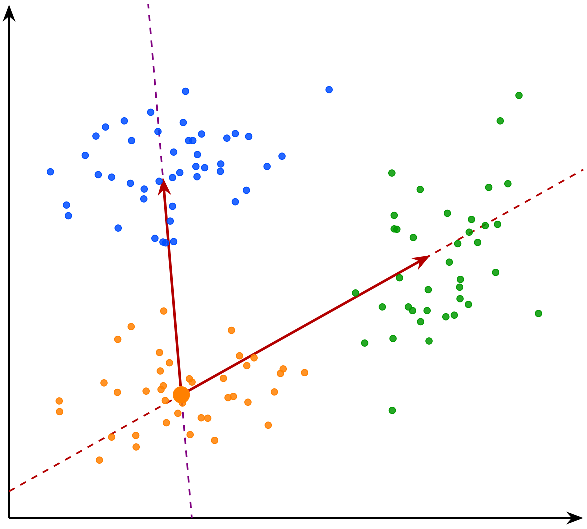


Figure 1.4: Interpretable directions in the latent space. The arrows represent semantic shifts from concept B toward concepts A and C. Moving along these directions in the latent space induces a coherent change in the model's behavior, suggesting that the transition between concepts is continuously encoded along these axes.

In many cases, the resulting change in the model's output varies smoothly with α, suggesting an approximately linear structure of the latent space with respect to certain semantic factors. This observation has led to the hypothesis that neural representations may organize concepts in a partially linear manner within the embedding space. Vectors of this form are referred to as interpretable directions, as they correspond to transformations that can be associated with specific semantic attributes. The objective of this line of work is therefore to identify directions in latent space that align with meaningful variations in the encoded concepts.

This phenomenon has been extensively studied in generative image models. As illustrated in Figure 1.5, controlled perturbations of latent representations along interpretable directions can induce specific semantic transformations, such as stroke thickness modification in digit images, head rotation in facial images, or texture and background changes in natural images [Voynov and Babenko, 2020]. These observations have led to the development of the field of latent space editing, whose objective is to manipulate generated content directly through modifications in latent space rather than through changes in the input prompt.

However, a central challenge remains: identifying such interpretable directions is far from trivial. Most existing approaches rely on auxiliary models that encode human-defined semantics, such as CLIP, to relate latent variations to textual concepts. Consequently, these methods inherit a limitation similar to that of probing techniques, since concept discovery depends on a pre-existing semantic framework defined by humans.

To overcome this limitation, a research direction known as unsupervised semantic discovery has emerged. Its goal is to uncover interpretable directions in latent spaces without explicit semantic supervision. In these approaches, one

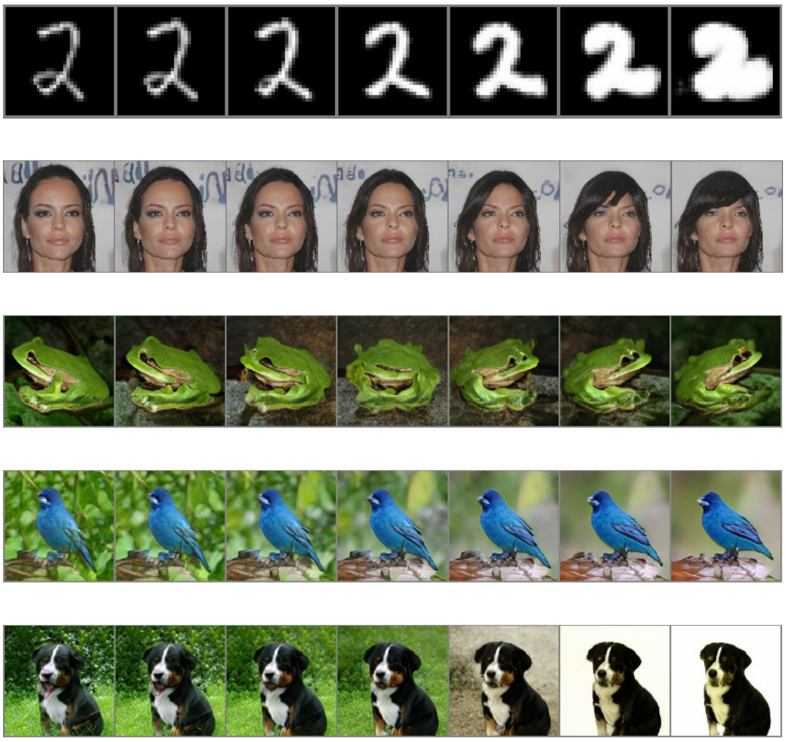
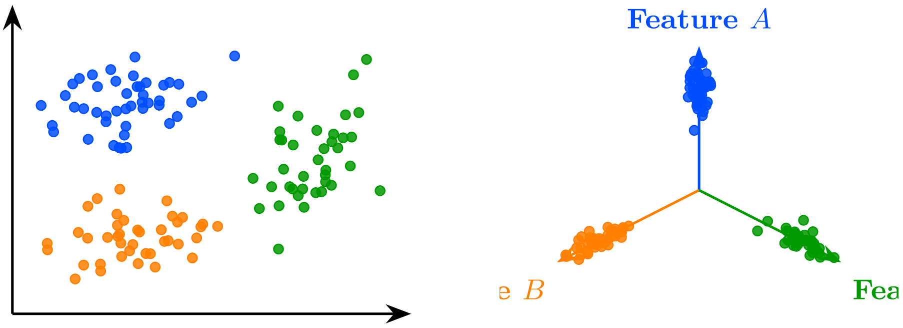


Figure 1.5: Unsupervised discovery of interpretable directions in the latent space of a generative model [Voynov and Babenko, 2020]. Each row corresponds to a distinct direction $\mathbf{v}_i$, and each column to an increasing value of α. The resulting transformations are semantically coherent and diverse: stroke thickness for digits, head rotation for faces, texture and background for natural images.

model applies perturbations along random directions in latent space, while a second model attempts to predict or recognize the applied transformation. If a stable correspondence emerges between the perturbation and its prediction, this suggests that the considered direction corresponds to a coherent structure in latent space.

In this framework, human intervention occurs only a posteriori, to verify whether the discovered directions indeed correspond to interpretable semantic variations. This approach enables the capture of representations that go beyond the mere presence or absence of a concept, including notions such as the intensity or degree of an attribute. As such, it follows a philosophy similar to causal probing, while also enabling explicit interventions to assess the causal role of latent representations.

However, despite these advances, several challenges remain. First, not all concepts encoded by neural networks appear to be linearly separable in latent space. While some directions align well with clear semantic factors, many others do not exhibit such a simple linear structure. In addition, latent representations often display a phenomenon known as entanglement, where multiple semantic attributes are mixed within the same directions of the representation space.



(a) Dense latent space **(b)** Sparse SAE space

Figure 1.6: Transformation from a dense latent space to a sparse SAE-style representation. On the left **(a)**, embeddings are entangled in the initial latent space. On the right **(b)**, the sparse representation aligns each concept with a dedicated feature axis.

Ideally, one would like each semantic factor to vary independently along a specific direction, without affecting other attributes. In practice, this is rarely the case. A single direction in latent space often influences multiple semantic features at once, meaning that modifying one aspect of the representation can unintentionally alter several others. This lack of separability indicates that semantic information is not cleanly disentangled in the representation space. Together with the difficulty of isolating independent factors, this observation has motivated a stronger hypothesis about how neural networks organize information: the superposition hypothesis.

1.4 Sparse Autoencoder

As discussed previously, approaches based on latent directions face inherent limitations. Empirically, only a restricted subset of concepts appears to admit a linear representation, and even these directions are often affected by significant entanglement with other concepts.

To explain this phenomenon, a complementary hypothesis has been introduced, extending the assumptions underlying the Concept Hypothesis (section 1.1). This hypothesis, known as the Superposition Hypothesis [Elhage et al., 2022], posits that concepts are preferentially represented in a linear fashion, with each concept ideally corresponding to a distinct direction in latent space. In this idealized setting, concept vectors would form an approximately orthogonal basis, where each direction independently controls a single factor of variation. Furthermore, only a small subset of the available concepts is expected to be active for a given representation. As a result, latent representations can be viewed as sparse combinations of concept directions. This idealized representation is referred to as a non-superposed representation and can be expressed as follows:

$$\mathbf{h} = \sum_{i=1}^{m} a_i \mathbf{v}_i, \qquad \forall i \neq j, \quad \mathbf{v}_i^\top \mathbf{v}_j \approx 0, \qquad \|\mathbf{a}\|_0 \ll m,$$

Figure 1.7: Example of a feature learned by a sparse autoencoder applied to a large language model [Templeton et al., 2024]. The highlighted tokens show the contexts in which the feature activates strongly, revealing a coherent semantic concept across English text, multiple languages, and images. This multimodal consistency suggests that the learned feature corresponds to a meaningful and generalizable concept, here related to the Golden Gate Bridge, rather than a spurious statistical pattern.

where:

- $\mathbf{h}$ denotes a latent representation,
- $\mathbf{v}_i$ represents the direction associated with concept c_i,
- a_i corresponds to the activation strength of concept c_i in the representation $\mathbf{h}$,
- $\|\mathbf{a}\|_0 \ll m$ indicates that only a small fraction of concepts are active for a given representation.

According to the Superposition Hypothesis, the reason why latent representations do not perfectly follow this idealized structure is that the representational capacity of neural networks is limited. For a given layer, the dimensionality of the representation space is finite, which constrains the number of concepts that can be encoded independently. As a result, the network cannot allocate a dedicated direction to every concept. Instead, multiple concepts are compressed into shared dimensions, a phenomenon referred to as superposition. This notion of superposition is closely related to the phenomenon of entanglement observed in latent space analyses. From the perspective both phenomena manifest themselves through the difficulty of isolating semantic factors into independent directions. However, while entanglement is an empirical observation about the structure of latent representations, superposition provides a theoretical explanation for why such mixing may emerge.

If this hypothesis holds, then any attempt to analyze latent representations will inevitably be affected by the phenomenon of superposition. Motivated by this view, and with the goal of recovering the idealized non-superposed, a large body of work has focused on developing methods to desuperpose latent representations. To this end, a common strategy consists in projecting latent representations into a higher-dimensional space while imposing additional constraints designed to recover a non-superposed representation. A widely used implementation of this idea is the sparse autoencoder, which can be defined as follows:

$$s = f_{\text{enc}}(h), \quad \hat{h} = f_{\text{dec}}(s)$$
$$\mathcal{L} = \|h - \hat{h}\|_2^2 + \lambda \|s\|_1$$

where:

- $f_{\text{enc}}(h)$ is the encoding function that projects the original representation space into a sparse latent space,
- $f_{\text{dec}}(s)$ is the decoding function that reconstructs the original representation from the sparse code,
- s is the sparse representation of h,
- $\|h - \hat{h}\|_2^2$ is the reconstruction loss ensuring that the original representation can be faithfully recovered,
- $\lambda\|s\|_1$ is the sparsity regularization term that encourages most components of s to be zero.

The method proves to be powerful when successfully applied to real language models. In these applications, the idea is to desuperpose latent representations using Sparse Autoencoders (SAEs) applied to LLM activations in order to recover a non-superposed representation. Once this representation is obtained, it has been shown that many concept vectors exhibit strong semantic properties. For instance, some features selectively activate on base64-encoded text or on Arabic script, and manually activating them can cause the model to generate outputs in the corresponding format.

With such representations, it becomes possible to perform interventions directly in latent space. A well-known example is the "Golden Gate Bridge" feature. A concept vector in the non-superposed space activates systematically when the input text or image refers to the Golden Gate Bridge. However, it is also possible to artificially activate this feature in a sparse representation even when the input does not contain any reference to it. After modifying the sparse representation, it is then projected back into the original activation space using the decoder function $f_{\text{dec}}(s)$. The model is then allowed to proceed with standard next-token generation.

Empirically, it is observed that even when the prompt is unrelated to the Golden Gate Bridge, the model's output shifts to discussing it. This provides evidence for the causal role of these learned features. This type of intervention has been used by Anthropic researchers to steer or suppress specific model behaviors, effectively enabling a form of behavior control through latent-space editing.

Despite their empirical success, sparse autoencoders exhibit several important limitations that question some of their underlying assumptions. hese limitations can be summarized as follows:

- **Non-identifiability of learned concepts.** The learned concepts are not guaranteed to be identifiable: different training runs may produce different decompositions of the same latent space, even when achieving comparable reconstruction performance. This raises questions about the stability and uniqueness of the learned representations.

- **Sparsity–semantics trade-off.** Empirical studies show that enforcing excessive sparsity can degrade the semantic quality of the learned features. In particular, when the sparsity coefficient is increased beyond a certain threshold, the resulting representations tend to lose meaningful semantic structure, suggesting a trade-off between sparsity and interpretability.
- **Failure of scaling to remove superposition.** The Superposition Hypothesis would suggest that increasing model capacity or latent dimensionality should systematically lead to more disentangled representations and improved performance. However, this is not consistently observed in practice. Even in very high-dimensional latent spaces, representations remain partially superposed, indicating that scaling alone does not fully resolve the entanglement of features.

These limitations suggest that sparse autoencoders, while powerful, do not fully resolve the problem of representational structure. Although empirical results are often convincing in practice, they do not necessarily validate the underlying theoretical assumptions.

From this perspective, probing methods, interpretable directions, and sparse autoencoders all rely on a shared assumption: that semantic structure can be recovered through linear or approximately linear decompositions of latent representations. While this assumption has led to significant empirical progress, it remains theoretically fragile. The following section therefore introduces a complementary approach that relaxes this constraint and does not require a globally linear feature decomposition of the latent space, namely deep clustering.

1.5 Clustering

The latent space analysis methods presented so far all rely, either explicitly or implicitly, on assumptions about the linear organization of concepts within the latent space. For probing methods, concepts must be at least partially linearly separable for simple probes to succeed. Interpretability through latent directions assumes that variations associated with a concept can be captured by moving along a linear axis of the latent space. Similarly, the superposition hypothesis assumes that concepts are represented through linear directions that become entangled because of representational capacity constraints. Although these assumptions have led to numerous empirical successes, their theoretical justification remains limited. More importantly, several observations suggest that the linearity hypothesis may not fully capture the structure of latent representations.

For instance, some concepts can only be recovered using non-linear probes. Likewise, methods based on interpretable directions typically identify only a limited number of semantic factors. Finally, in sparse autoencoder approaches, enforcing excessive sparsity often degrades reconstruction quality and interpretability, suggesting that the underlying representation may not be perfectly described by a sparse linear decomposition.

Taken together, these observations indicate that semantic concepts may not always be organized according to a simple linear structure. This motivates the exploration of alternative approaches that rely on weaker assumptions regarding the geometry of latent spaces. One such approach is clustering-based analysis.

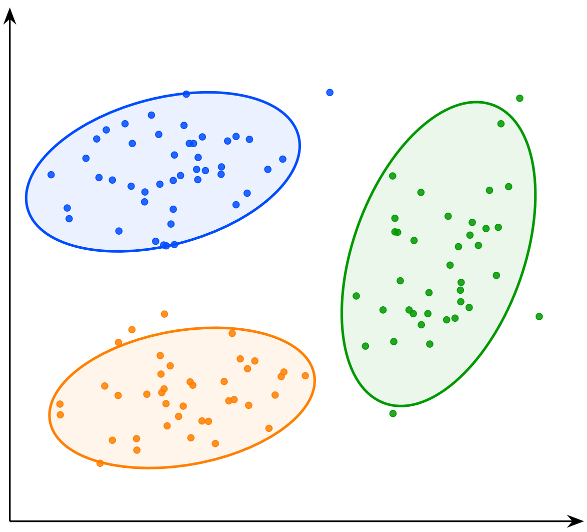


Figure 1.8: Visualization of the latent space structure. The points represent embeddings associated with concepts A, B, and C. The ellipses indicate the approximate extent of each group in the latent space and highlight their geometric separation.

Rather than assuming that concepts correspond to directions or linear subspaces, clustering methods only assume that semantically similar inputs produce nearby latent representations. Conversely, inputs that differ significantly from a semantic perspective are expected to be located farther apart in the latent space.

Under this assumption, the discovery of concepts can be reformulated as the identification of groups of embeddings that occupy the same region of the latent space. Concepts are therefore no longer associated with directions, but with clusters of neighboring representations. This idea forms the basis of the research field known as deep clustering. In these approaches, a collection of inputs is first projected into a latent space using a neural network. The resulting embeddings form a point cloud on which standard clustering algorithms can be applied. The obtained clusters are then interpreted as representing distinct semantic concepts.

This methodology enables the discovery of semantic structure in the latent space without requiring concepts to be linearly encoded. As a result, it can reveal forms of organization that would remain inaccessible to methods based solely on linear separability or interpretable directions. Deep clustering has been particularly successful in unsupervised image classification.

Typically, an autoencoder is composed of an encoder f_{enc} and a decoder f_{dec}. The encoder maps an input $x^{(i)} \in \mathcal{X}$ to a low-dimensional latent representation $z^{(i)} \in \mathbb{R}^d$:

$$z^{(i)} = f_{\text{enc}}(x^{(i)}), \quad d \ll \dim(\mathcal{X}).$$

To ensure that this compressed representation does not discard essential information, the decoder reconstructs the input from the latent space:

$$\hat{x}^{(i)} = f_{\text{dec}}(z^{(i)}) = f_{\text{dec}}(f_{\text{enc}}(x^{(i)})).$$

The model is trained to minimize the reconstruction error between the original

Figure 1.9: Clusters discovered by a deep clustering model applied to natural images [?]. Each row contains the ten highest-scoring elements of a single cluster. The visual coherence within each row, grouping animals, vehicles, and aircraft into distinct clusters, suggests that the learned latent representations capture semantically meaningful structure without any explicit supervision.

input and its reconstruction, typically using a mean squared error (MSE) loss:

$$\mathcal{L}_{\text{rec}} = \frac{1}{N}\sum_{i=1}^{N} \left\| x^{(i)} - \hat{x}^{(i)} \right\|_2^2 .$$

This objective encourages the latent space to preserve the most informative features of the input while enforcing a compact representation.

Clustering algorithms are then applied to the learned embeddings.

Let $\mathcal{D} = \{x^{(i)}\}_{i=1}^{N}$ denote a dataset of inputs. Through the encoder f_{enc}, this dataset induces a point cloud in the latent space:

$$\mathcal{Z} = \{z^{(i)} \in \mathbb{R}^d \mid z^{(i)} = f_{\text{enc}}(x^{(i)}),\ x^{(i)} \in \mathcal{D}\}.$$

Clustering algorithms are then applied to this set $\mathcal{Z}$ of embeddings in order to identify groups of semantically similar representations.

An example of this methodology can be observed in Figure 1.9, where the approach is applied to the STL-10 dataset using k-means clustering with a number of clusters arbitrarily fixed to $K = 10$. For each cluster, the corresponding line shows the 10 samples closest to the associated centroid in the latent space.

We observe that the method captures coherent semantic categories such as animals, vehicles, and flying entities. This result is particularly striking given that the only available information to the system consists of raw images. It suggests that the compression process induces a latent space in which semantically meaningful structures naturally emerge, and that these structures correspond to high-level concepts.

More advanced approaches combine representation learning and clustering objectives within a single optimization process. In this setting, additional loss terms encourage embeddings to organize themselves around cluster centroids during training. This often produces latent spaces that are easier to analyze and improves the quality of the discovered semantic structure.

Despite its effectiveness, clustering-based analysis also presents important limitations. First, as with most unsupervised semantic discovery techniques, the interpretation of the resulting clusters remains partially subjective. While benchmark datasets often provide class labels that facilitate evaluation, there is generally no guarantee that the discovered clusters correspond to the categories that humans consider most relevant. Alternative semantic groupings may be equally valid.

Second, clustering methods are highly sensitive to design choices. The number of clusters, the distance metric, and the clustering algorithm itself can significantly influence the results. Selecting these parameters is often challenging and may require substantial empirical tuning. Finally, semantic concepts do not necessarily form well-separated clusters in latent space. Some concepts may overlap, exhibit hierarchical relationships, or vary continuously rather than discretely. In such cases, a clustering-based description may provide only a partial view of the underlying representational structure.

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